



Prevention Without Proof

Did national suicide-prevention strategies reduce suicide?

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40s

Somewhere on earth, about every forty seconds, a person decides the rest of life is not worth the rest of life.



More than war, more than homicide

More than **700,000** suicides a year.

More than war.

More than homicide.

More than most diseases we built ministries
to defeat.

The number is stubborn

It has barely moved in a generation.



The world answered with a document

Finland, 1986–92: investigate every suicide, build a national program.

Then everyone copied it: Norway and Sweden 1995, France 2000, the UK 2002, Ireland 2005.

By 2018 the WHO urges *every* member state to adopt one.

The premise

We decided, collectively, that the **strategy document** is the cure.

The only question that matters

Does a country that adopts a national strategy
bury **fewer** of its citizens than it otherwise would have?

Empirical. Answerable in principle. Almost entirely unanswered in practice.



My answer, stated up front

From the cross-national record, **we cannot tell** whether these strategies have saved a single life.

That should frighten the skeptic and the believer alike.

The famous answer

A horizontal bar spanning the width of the slide, divided into three equal segments. The left segment is green, the middle segment is white, and the right segment is red.

And the method underneath it



The canonical finding looked protective

Matsubayashi & Ueda (2011): across 21 OECD countries, those with national programs saw suicide **fall** relative to those without.

It became the evidence base for a global policy.

The engine

A two-way fixed-effects comparison of adopters vs. non-adopters.



Two-way fixed effects misleads under staggered timing

When units adopt at different dates and effects are heterogeneous, TWFE uses **already-treated units as controls** for newly-treated ones.

- The weights can be negative.
- The estimate can fall outside the range of every true effect.
- It can return the **wrong sign** for mechanical reasons.

Goodman-Bacon (2021); de Chaisemartin & D'Haultfœuille (2020).

The data

All real — nothing simulated

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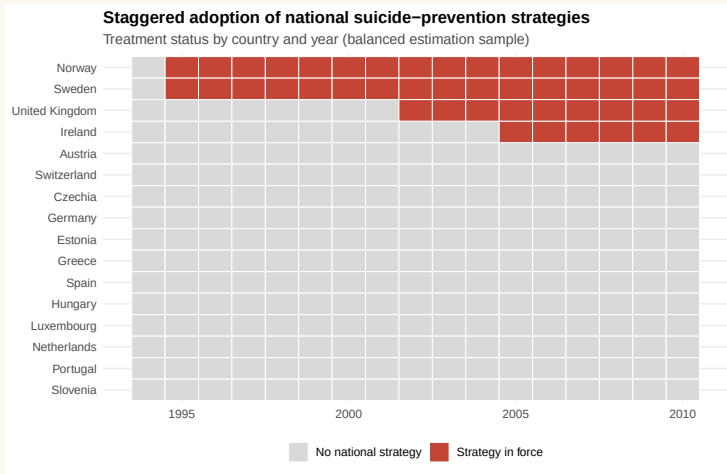
Three real sources, one country-year panel

- **Outcome:** Eurostat age-standardised suicide rate (ICD-10 X60–X84, Y87.0), 1994–2010.
- **Treatment:** year a government adopted a stand-alone national strategy (WHO 2018; Lewitzka et al. 2019).
- **Covariate:** World Bank unemployment — the business cycle.

Verified independently in R, Python, and Stata: identical samples and means.

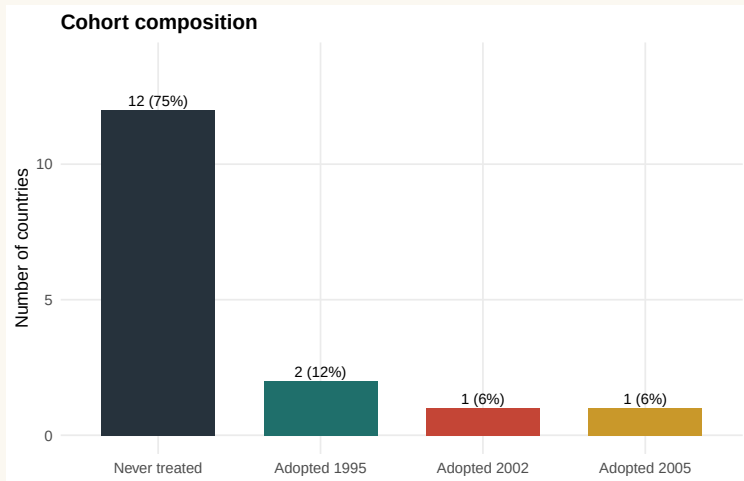


The natural experiment is staggered — and thin



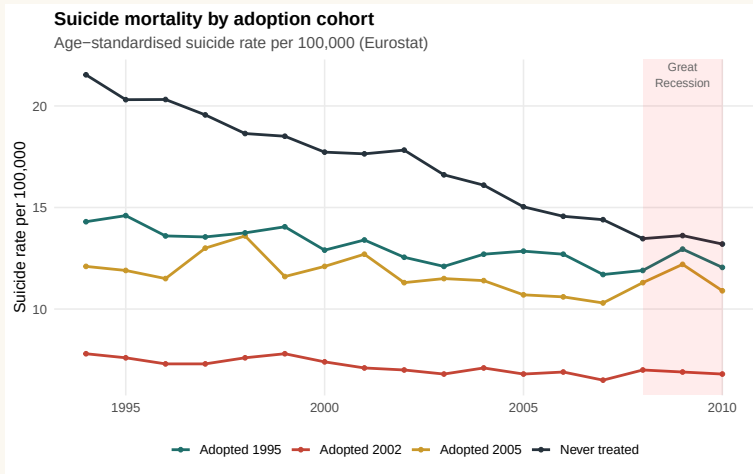


Four treated countries — I will not pretend otherwise





Treated countries start lower; all bend up in the recession



The design

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What we are trying to measure



The estimand: a weighted average effect on the treated

For each adoption cohort g and year t , compare the change in suicide against not-yet- and never-treated countries; aggregate, weighting cohorts by size.

$$\theta = \sum_g w_g \overline{ATT}(g), \quad w_g \propto \# \text{treated countries in } g.$$

Callaway & Sant'Anna (2021), under *conditional* parallel trends.



The covariate an expert would insist on

Recessions raise suicide. Recoveries lower it.

And Europe's recessions are not independent of its strategies.

So we condition on **unemployment** — the fast-moving confounder that can wear a policy's clothes (Ruhm 2000; Reeves et al. 2012).



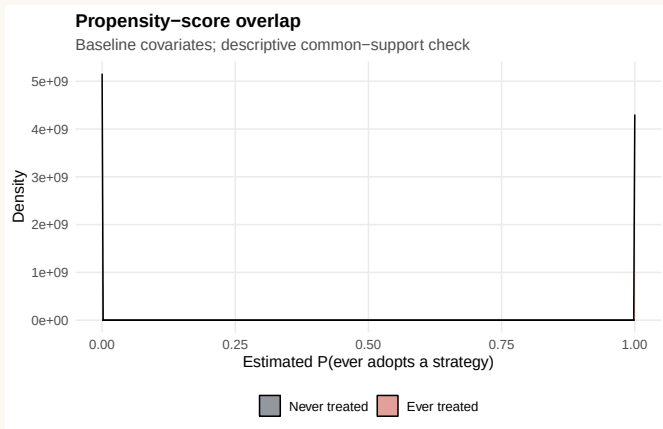
And unemployment is genuinely imbalanced

Baseline (1994)	Treated	Control	Std. diff.
Suicide rate (per 100k)	12.1	21.5	-0.99
Unemployment (%)	9.8	8.2	+0.33
Share aged 65+ (%)	15.2	14.0	+0.63
Median age (years)	35.2	36.2	-0.42

Imbens & Rubin (2015): $|\text{std. diff}| > 0.25$ is imbalanced. Unemployment clears it — conditioning is necessary, not cosmetic.



There is no common support to weight on



Perfect separation $\Rightarrow$ regression adjustment, not inverse-propensity weighting.



Sixteen clusters: trust the placebo, not the t -statistic

Four treated countries, twelve controls $\approx$ sixteen clusters.

Cluster-robust and bootstrap standard errors need many-cluster asymptotics this design does not have.

So the inferential weight is carried by randomization inference

How often does *fake* adoption, assigned at random among the never-treated, produce an estimate this large?

The naive answer

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And why we must discard it



Differencing gives a positive, wrong-signed number

+2.7 to +3.2

more suicides per 100,000 after adoption

Callaway–Sant’Anna, Borusyak et al., Sun–Abraham — they agree on the number. None of them is identified here.

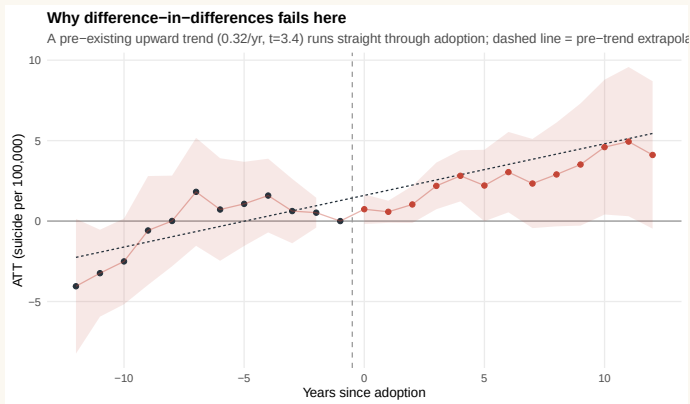
Why difference-in-differences is the wrong tool

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The groups differ in levels *and* trends



A trend runs straight through adoption



Pre-trend slope $+0.32/\text{yr}$ ($t = 3.4$, $p = .001$). The dashed extrapolation runs through the post-adoption points — no break at treatment.



When groups differ in levels and trends, do not difference

- **Levels** differ: every covariate is imbalanced (Imbens–Rubin > 0.25).
- **Trends** differ: a significant upward slope predates *and* survives adoption.

Differencing extrapolates the gap — it does not remove it.

Doudchenko & Imbens (2016): match on the level *path* instead. That is synthetic control.

Synthetic control

Match on the level path — for all four

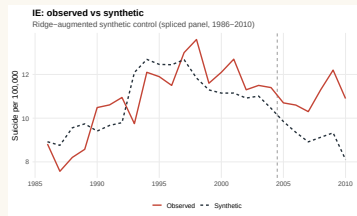
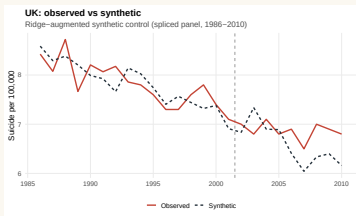
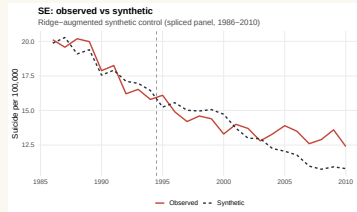
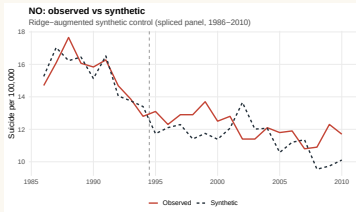
Build each adopter a synthetic twin

For each treated country: a weighted average of the 13 never-adopters that reproduces its **pre-adoption suicide path** (Abadie 2010), bias-corrected by ridge augmentation (Ben-Michael et al. 2021).

The WHO splice (from 1986) gives Norway & Sweden 9 pre-years, the UK 16, Ireland 19.

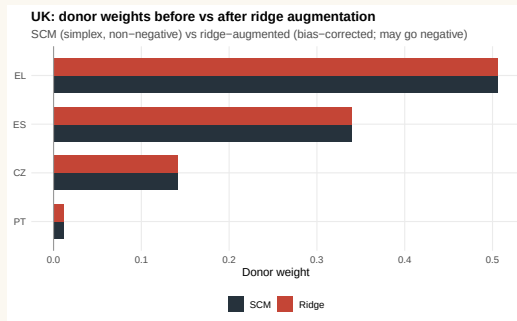
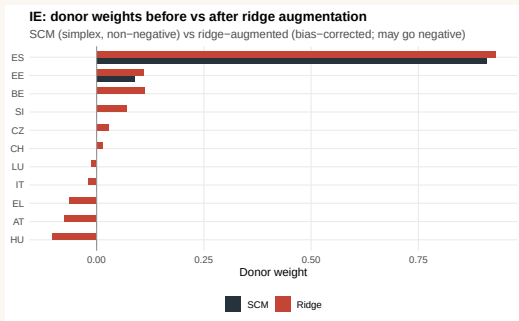


The synthetic twins track all four before adoption





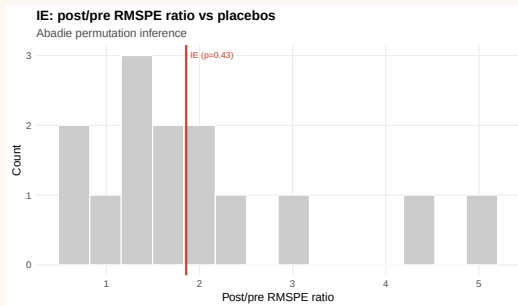
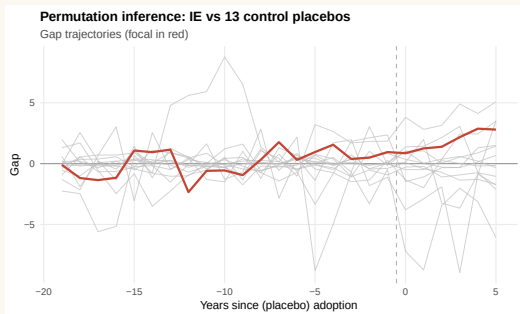
Donor weights, before and after ridge augmentation



Pure SCM (simplex, non-negative) vs ridge-augmented (bias-corrected). Ireland's poorer raw fit pushes some weights negative; the UK's near-perfect fit barely moves.



Permutation inference: every adopter is ordinary



Permutation p — Norway 0.71, Sweden 0.71, UK 0.79, Ireland 0.43 — all in the body of the placebo distribution.



Synthetic difference-in-differences agrees, by cohort

Adoption cohort	SDID estimate	s.e.
1995 (Norway, Sweden)	+0.80	4.20
2002 (United Kingdom)	+1.24	3.96
2005 (Ireland)	+0.74	1.44

Arkhangelsky et al. (2021). Every estimate smaller than its own standard error.



The appropriate method finds nothing

Four countries. Three methods. One verdict.

No detectable effect on suicide — and no power to rule one out.

Devil's advocate

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The strongest objection



“You are explaining away an inconvenient result”

The skeptic

You got a positive estimate and then argued it away with pre-trends and a recession. Convenient.

The response

I am *not* claiming the recession explains the result — that would identify, from a panel I called uninformative, an effect I have no business identifying. The claim is weaker and more durable: these forces are present, the design cannot separate them, and a number that confounds all three measures none.

What this means

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For the question, and for the Commission



Not that strategies fail — that we cannot see

The finding is not that prevention is worthless.
It is that the observational cross-section is **structurally incapable**
of revealing its worth.

A policy can be both worth keeping and unproven. Honesty means holding both.



We built a continent of strategies with no evaluation inside

No staggered randomization. No phased regional rollout with administrative mortality data. No pre-registered evaluation attached to a single national plan.

Every country switched at once, at the level where the confounders live, in response to the outcome itself.



For the Commission: design the evaluation before the policy

- Build the counterfactual **in**, not reconstruct it afterward.
- Evaluate **below the nation** — regions, phased rollout.
- Pre-register the mortality outcome before deployment.

The alternative is to keep spending moral and fiscal capital on faith.

Italy never wrote the document

The country in which we sit is a documented **never-adopter**.

A large European nation that never wrote the strategy —
the silent control against which the writers might have been judged,
had the data let us judge them.

We have been making life-and-death policy

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in a darkness we mistook for evidence.